(ii)

$$\overset{+3}{C_2}O_4^{2-} \longrightarrow \overset{+4}{C}O_2 \quad \text{..... (1)}$$

$$\overset{+6}{Cr_2}O_7^{2-} \longrightarrow Cr^{3+} \quad \text{..... (2)}$$

(1) $\Rightarrow$ $C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^- \quad$ ..... (3)

$Cr_2O_7^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O \quad$ ..... (4)

(3) $\times$ 3 $\Rightarrow$ $3C_2O_4^{2-} \longrightarrow 6CO_2 + 6e^- \quad$ ..... (5)

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H^2O \quad$ ..... (4)

$$Cr_2O_7^{2-} + 3C_2O_4^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 6CO_2 + 7H_2O$$

(iii) **அரை வினைகள்**

$S_2O_3^{2-} \longrightarrow S_4O_6^{2-} \quad$ ..... (1)

$\Rightarrow$ $I_2 \longrightarrow I^- \quad$ ..... (2)

(1) $\Rightarrow$ $2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2e^- \quad$ ..... (3)

(2) $\Rightarrow$ $I_2 + 2e^- \longrightarrow 2I^- \quad$ ..... (4)

(3) + (4) $\Rightarrow$ $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$

(iv) **அரை வினைகள்**

$\overset{0}{Z}n \longrightarrow Zn^{2+} \quad$ ..... (1)

$\overset{+5}{N}O_3^- \longrightarrow \overset{+2}{N}O \quad$ ..... (2)

(1) $\Rightarrow$ $Zn \longrightarrow Zn^{2+} + 2e^- \quad$ ..... (3)

(2) $\Rightarrow$ $NO_3^- + 3e^- + 4H^+ \longrightarrow NO + 2H_2O \quad$ ..... (4)

(3) $\times$ 3 $\Rightarrow$ $3Zn \longrightarrow 3Zn^{2+} + 6e^- \quad$ ..... (5)

(4) $\times$ 2 $\Rightarrow$ $2NO_3^- + 6e^- + 8H^+ \longrightarrow 2NO + 4H_2O \quad$ ..... (6)

$$3Zn + 2NO_3^- + 8H^+ \longrightarrow 3Zn^{2+} + 2NO + 4H_2O$$



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